

Vigashini S

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Profile

Machine Learning practitioner with experience in developing intelligent, data-driven solutions and AI-based applications. Strong understanding of core ML concepts including model training, evaluation, and optimization, along with a solid foundation in algorithms and system design. Passionate about applying machine learning techniques to solve real-world problems and build scalable systems.

Education

KPR Institute of Engineering and Technology

Sept 2023 – May 2027

BE Computer Science and Engineering

CGPA: 9.09/10.0

Sakthi Vigneswara Higher Secondary School

March 2023

HSC

Percentage: 98.5

Skills

- Programming: Python, Java, JavaScript
- Machine Learning: Supervised Unsupervised Learning, Model Training, Evaluation
- Data Handling: NumPy, Pandas, Data Preprocessing
- Databases: SQL, MongoDB
- Concepts: Data Structures, OOP, DBMS, Operating Systems
- Tools: Git, GitHub, Docker, AWS Fundamentals

Projects

Multi-Agent AI Task Executor

- Developed a multi-agent system with Planner, Executor, and Validator agents for automated task execution.
- Built backend orchestration for system control, APIs, and real-time operations.
- Implemented memory modules and feedback loop for improved decision-making.
- Tech Stack: Python, FastAPI, REST APIs, LLM APIs, Docker, SQLite | **GitHub: Link**

OverloadShield: Proactive System Failure Prevention Using AI

- A real-time system monitoring platform for continuous metric collection and overload detection.
- Implemented Gradient Boosting forecasting and Isolation Forest anomaly detection to identify system failures and abnormal patterns. Built FastAPI backend, Streamlit dashboard, and automated severity-based alerting system for real-time monitoring and response.
- Tech Stack: Python, FastAPI, Streamlit, Scikit-learn, Pandas, NumPy, psutil, REST APIs | **GitHub: Link**

AlertDrive AI: Real-Time Driver Drowsiness Detection System

- Developed a Driver Drowsiness Detection System using computer vision to monitor eye activity and detect fatigue in real time. Implemented face and eye detection using Haar Cascades with frame-based logic for blink detection and drowsiness alert triggering. Built a real-time monitoring interface with alarm system and CSV-based logging to track driver alertness and activity patterns.
- Tech Stack: Python, OpenCV, Pygame, Haar Cascade, CSV File Handling | **GitHub: Link**

Certifications

- Cisco: CCNA Introduction to Networks
- Coursera: Python for Data Science and AI (IBM)
- MongoDB for SQL Experts
- Infosys: Introduction to Cloud Computing

Internship

Tech Octanet Services Pvt Ltd – Java Developer Intern

- Developed backend modules using Java and collaborated using Git.
- Participated in debugging and improving application performance.
- Gained exposure to software development lifecycle and clean coding practices.

Achievements

- Solved 200+ problems on LeetCode, strengthening problem-solving and DSA skills.
- Earned Google Cloud badge for developing GenAI applications using Gemini.
- Secured 1st place in a technical paper presentation.